Movement and Control of Biomimetic Humanoid Robots

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November 6, 2026

Portland State UNIVERSITY

AGILE & ADAPTIVE ROBOTICS LABORATORY

C³NS Communication, Coordination, and Control in Neuromechanical Systems

NeuroNex

Robots on warehouse floors face different challenges from robots on rubble



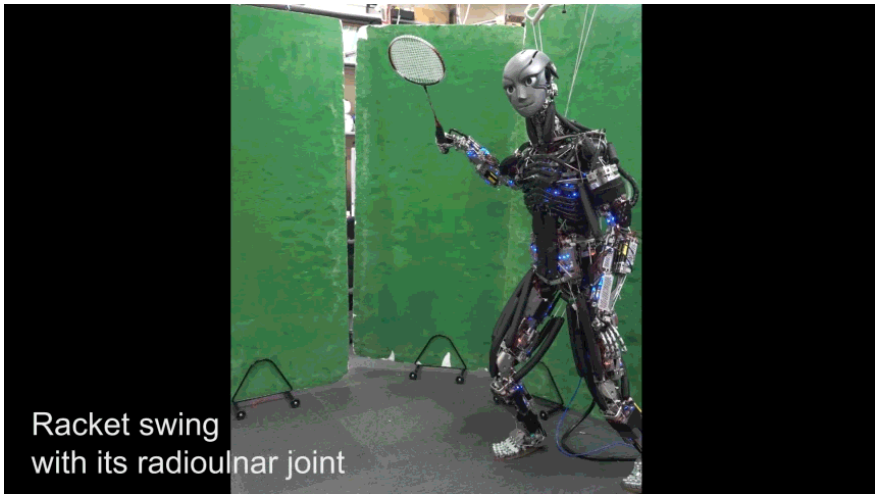
Predictable contact



Changing contact and balance

Amazon Robotics; DARPA Robotics Challenge (2015)

Biomimetic robots let us test mechanics and neural control



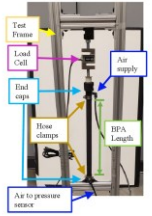


Biological mechanisms

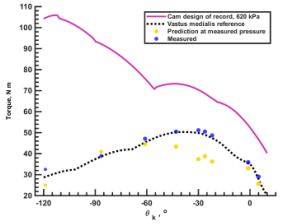
Controlled robot experiments

Asano et al. (2016); Bolen and Hunt (2019); Dissertation Ch. 1

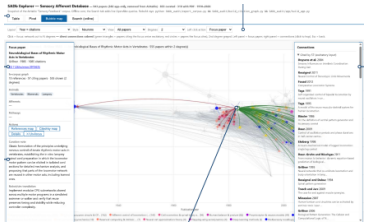
Four knowledge and tool gaps for biomimetic robot research



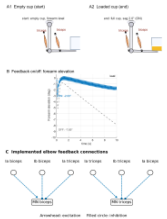
BPA characterization
How does length change force?



Joint design factors
What torque reaches the joint?



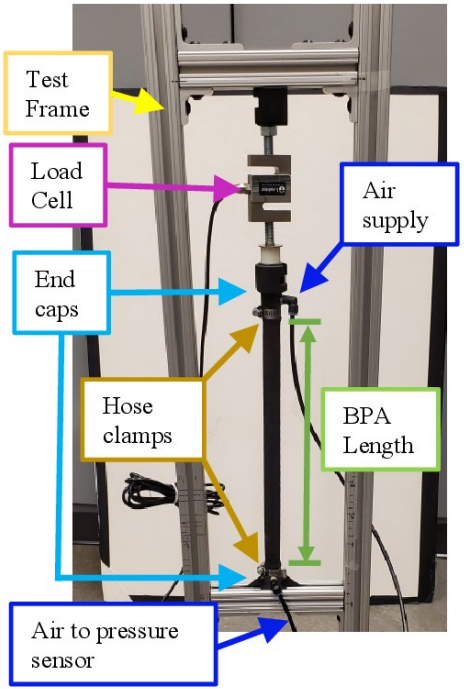
SADb
Which sensory evidence applies?



SNS tools
Which pathway causes the behavior?

Dissertation Chs. 3–6

BPA force depends on length, pressure, and contraction



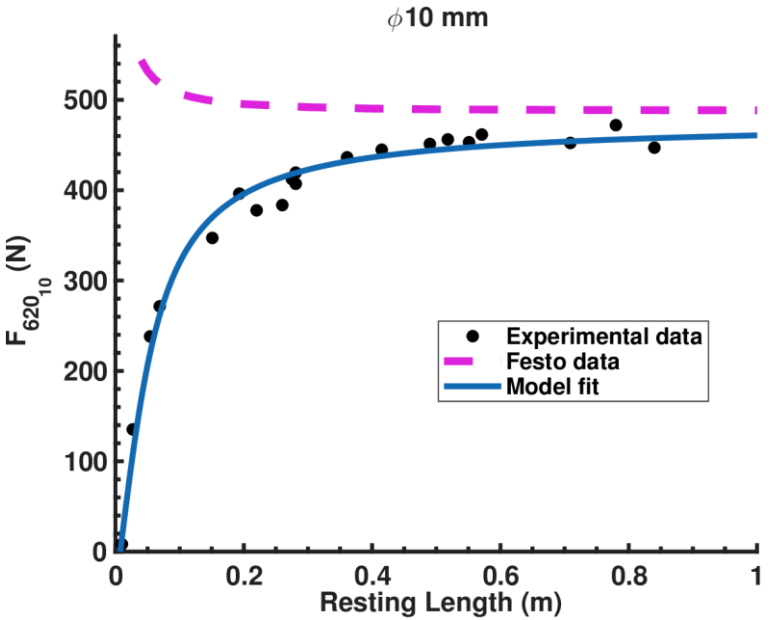
Contracted

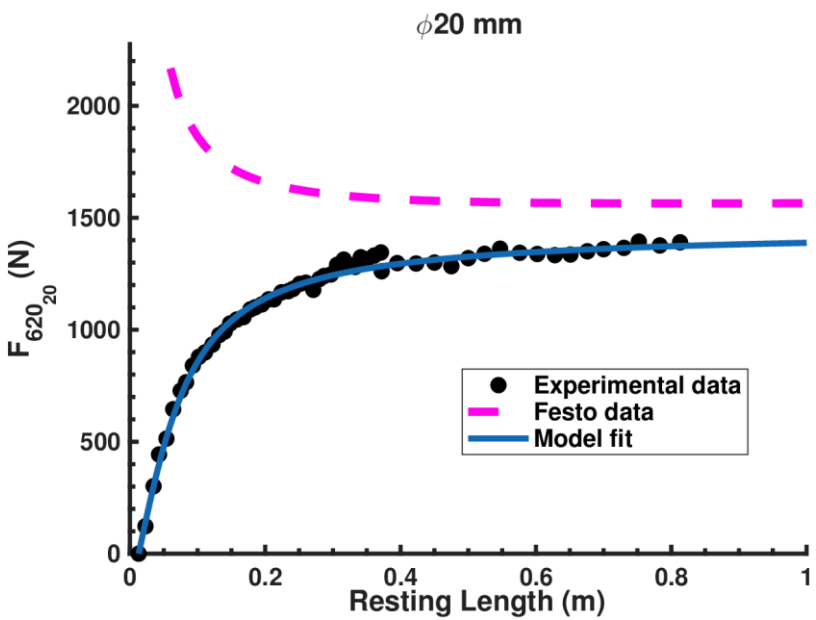


Uncontracted

Bolen et al., Actuators 15(5):230 · Hunt, INRC (2019), slide 49

Shorter BPAs deliver less maximum isometric force



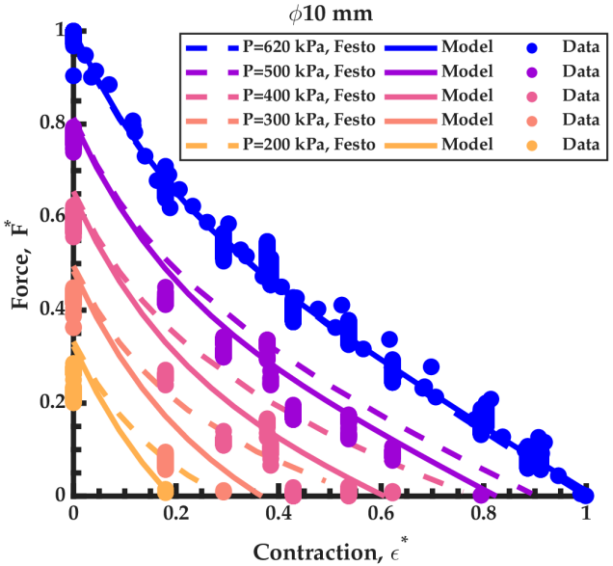


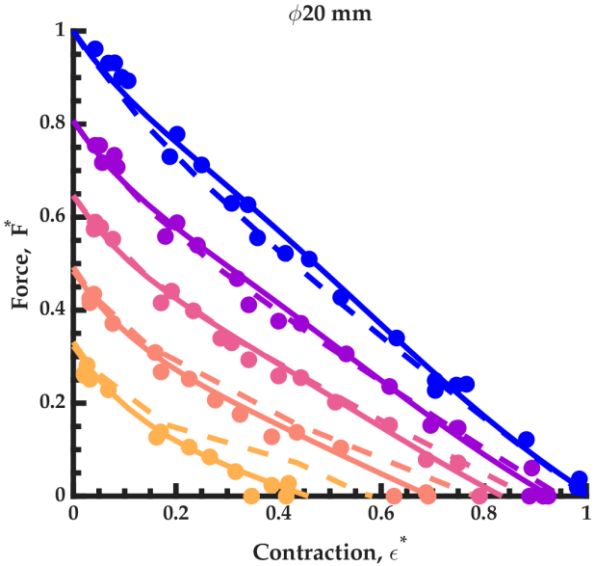
Resting length changes the force available to a robot design

Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

Resting-length scaling completes the normalized force model

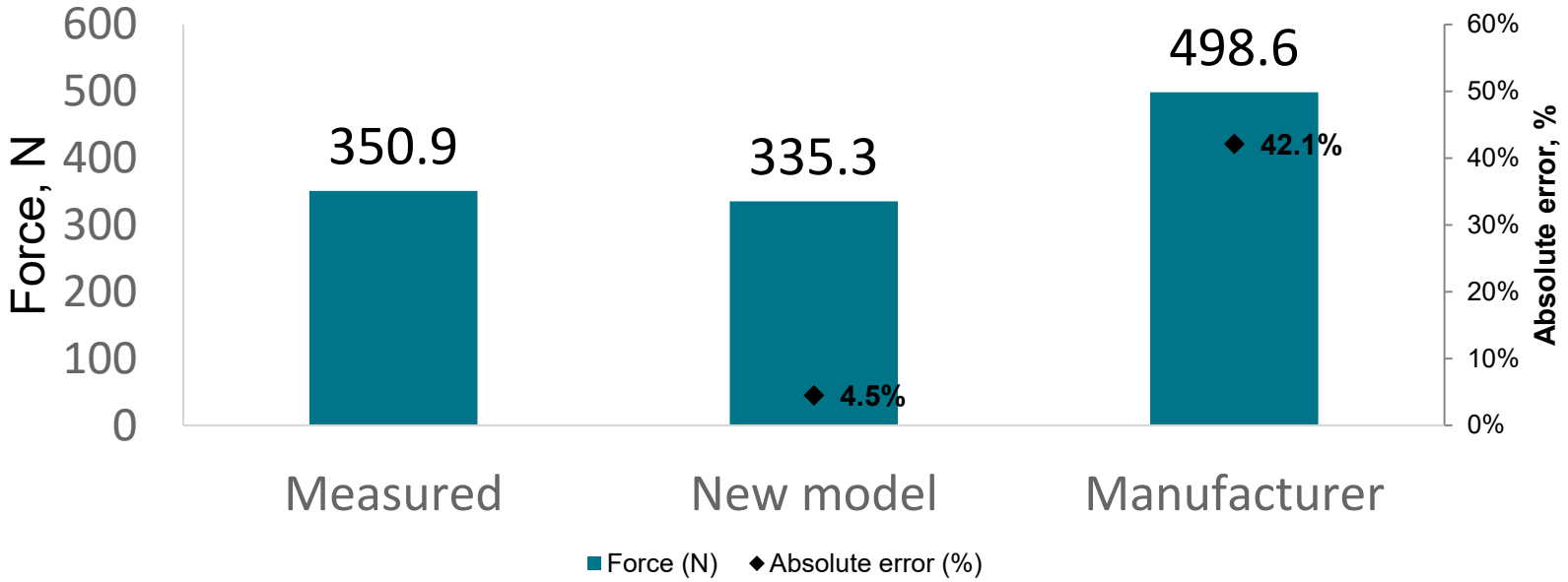
$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0)$$





Bolen et al., Actuators 15(5):230 · Dissertation Ch. 3

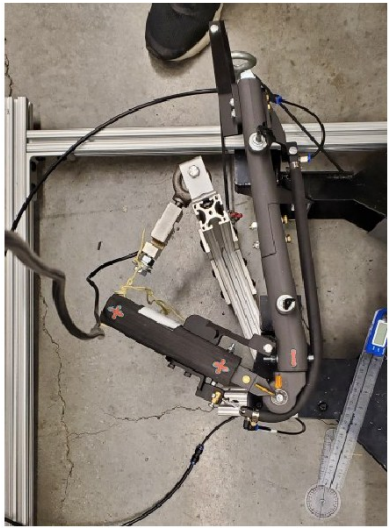
At 112 mm, the new model reduces force error to 4.5%



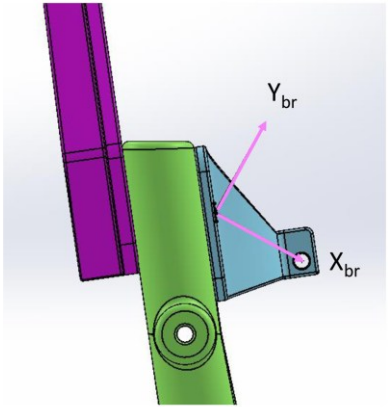
10 mm BPA · 112 mm resting length · 620 kPa

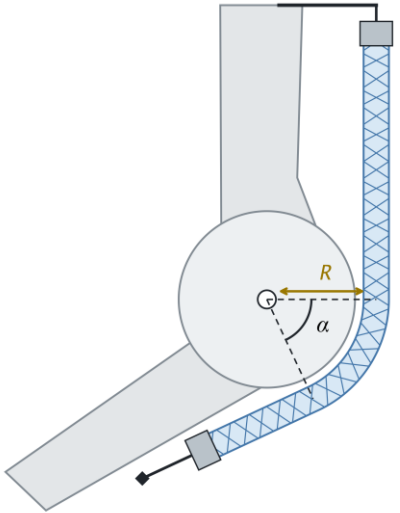
Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

Length offsets, compliance, and wrapping change predicted joint torque



(B)



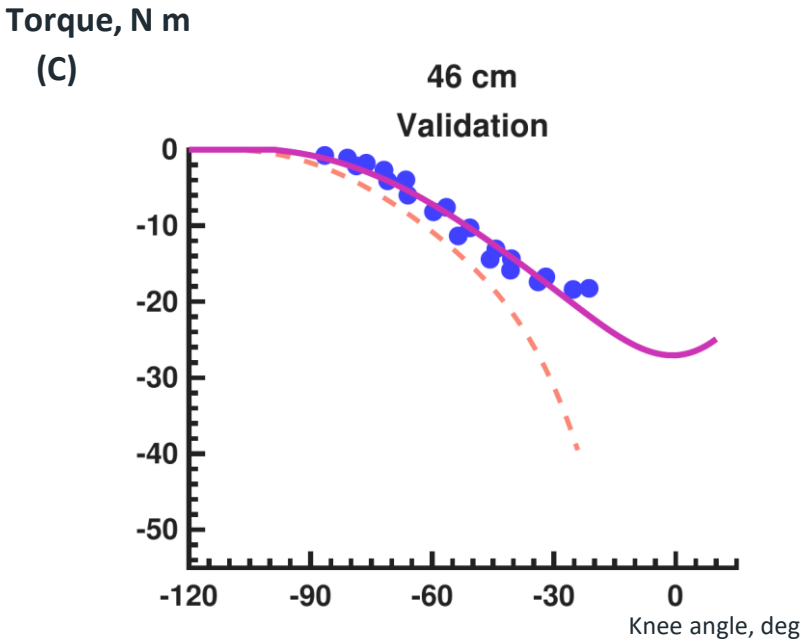


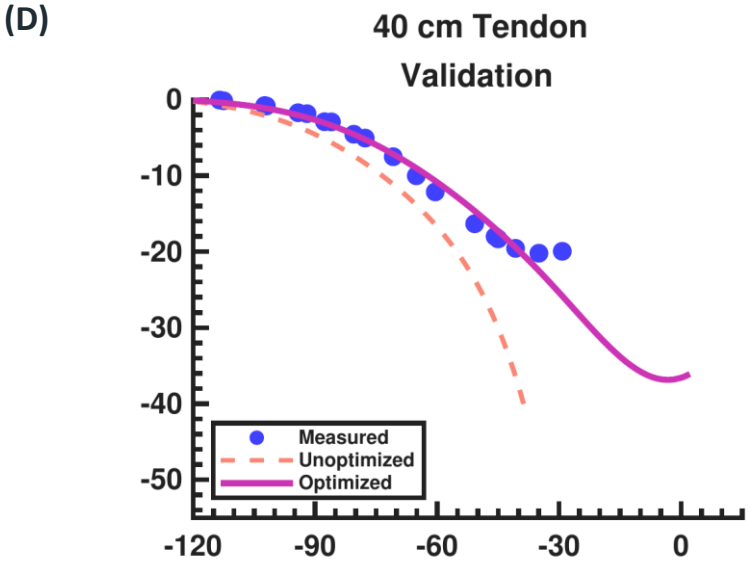
χ_0
Unloaded length offset

χ_1, χ_2
Effective system stiffness

χ_3
Wrapping length loss

Design corrections reduce error on held-out knee tests

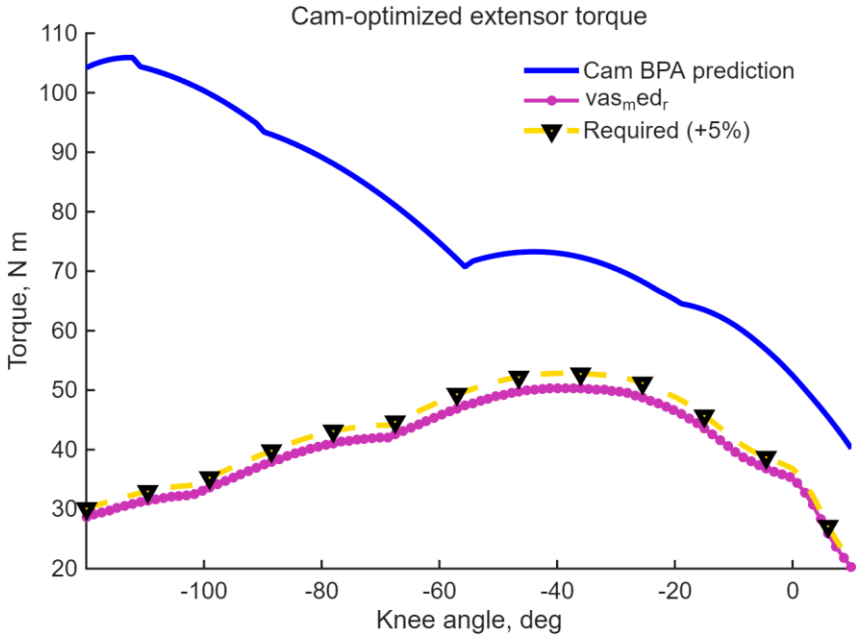




Held-out mean RMSE: 7.588 to 1.490 N m

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

The corrected model predicts a cam route with torque headroom



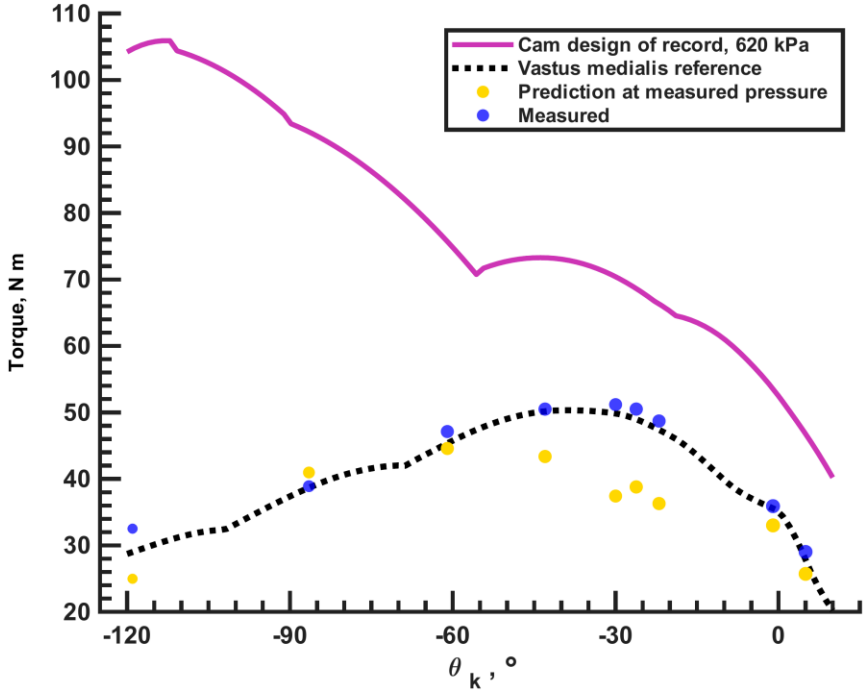
+30.83% above required envelope · 269–507 kPa predicted pressure

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

All nine cam-route measurements exceed the selected human target



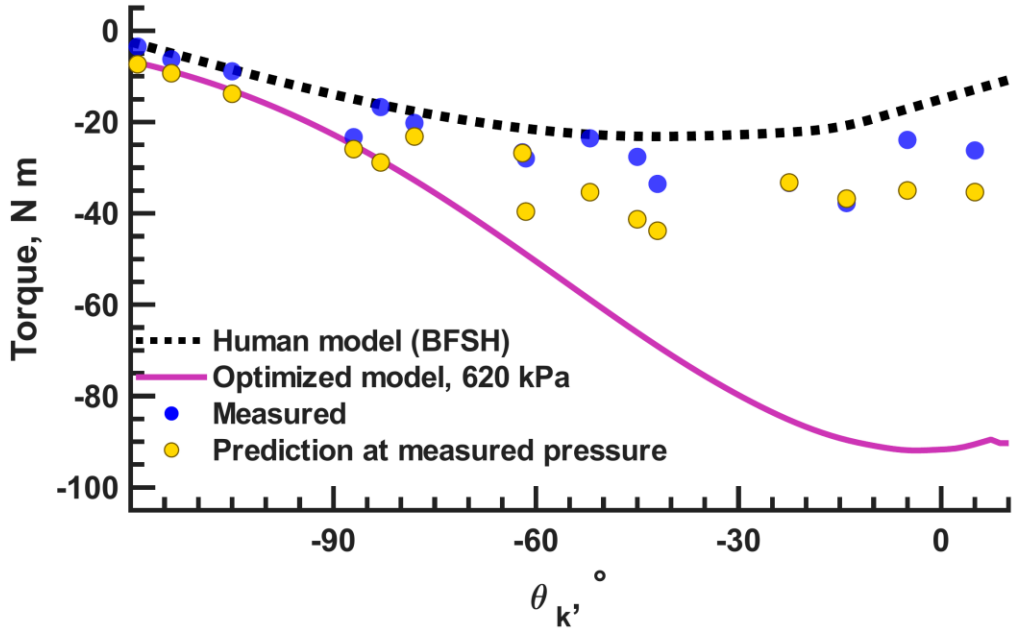
Two 20 mm BPAs



9 of 9 human targets met

Dissertation Ch. 4 · Physical extensor tests

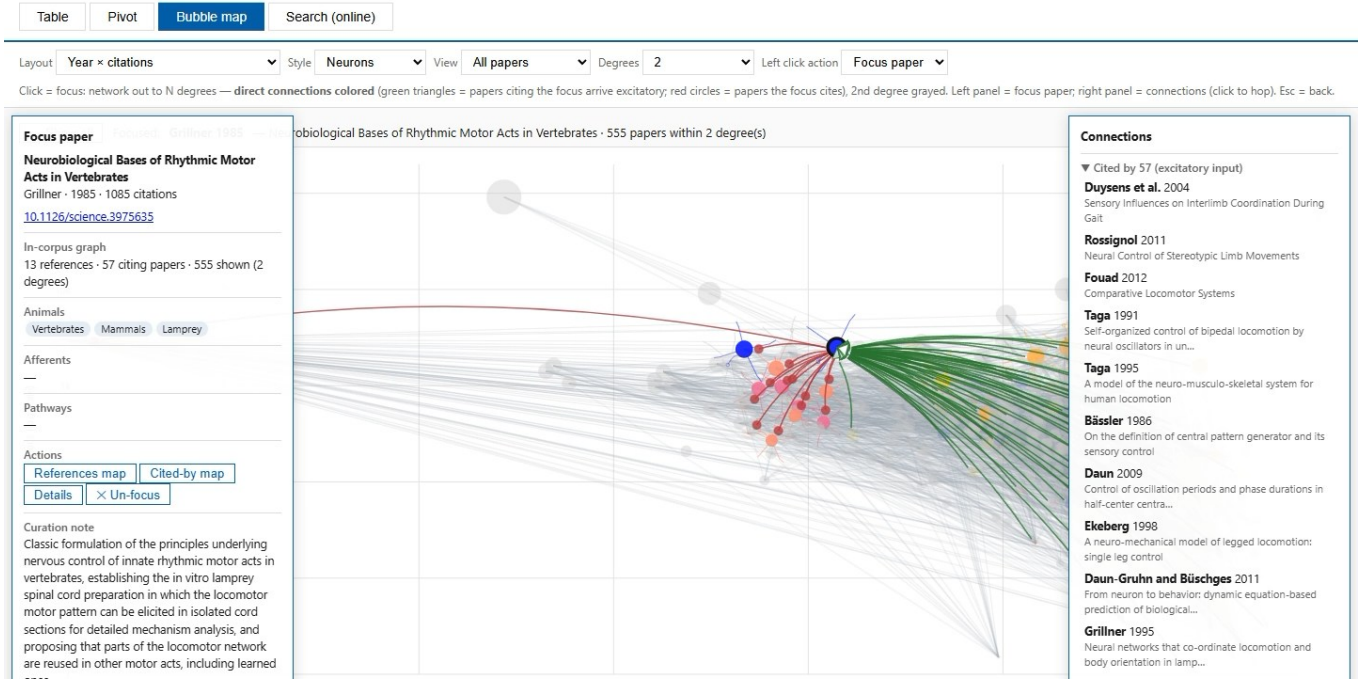
All fifteen flexor measurements meet the human target



15 of 15 targets met · 226.5–638 kPa · negative torque denotes flexion

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

SADb makes sensory evidence searchable and traceable



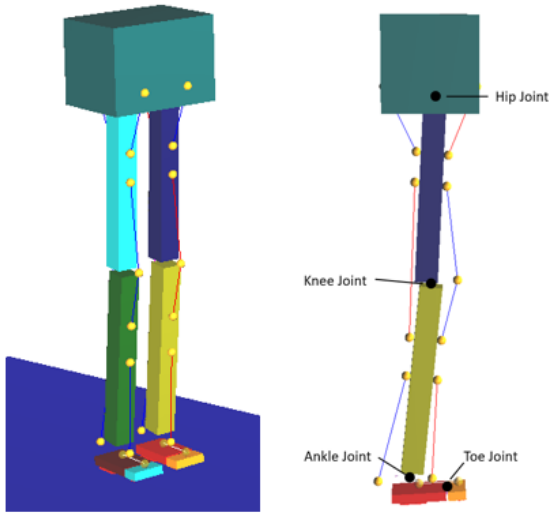
Open offline SADb

SADb Explorer · Offline snapshot: 1,393 records

Search · inspect · follow · compare

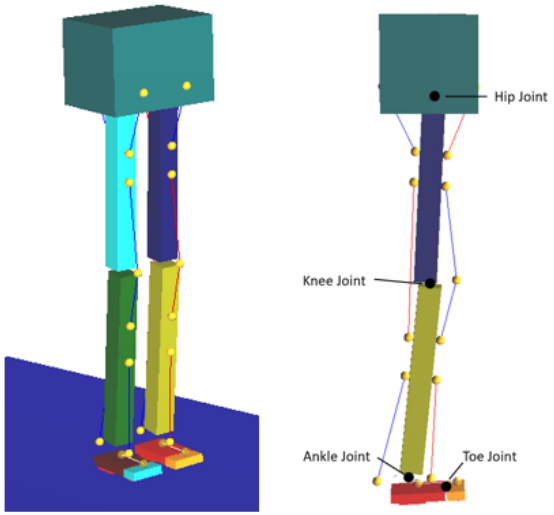
Suspended leg motion does not establish weight-bearing walking

Suspended



Native recording pending

Released

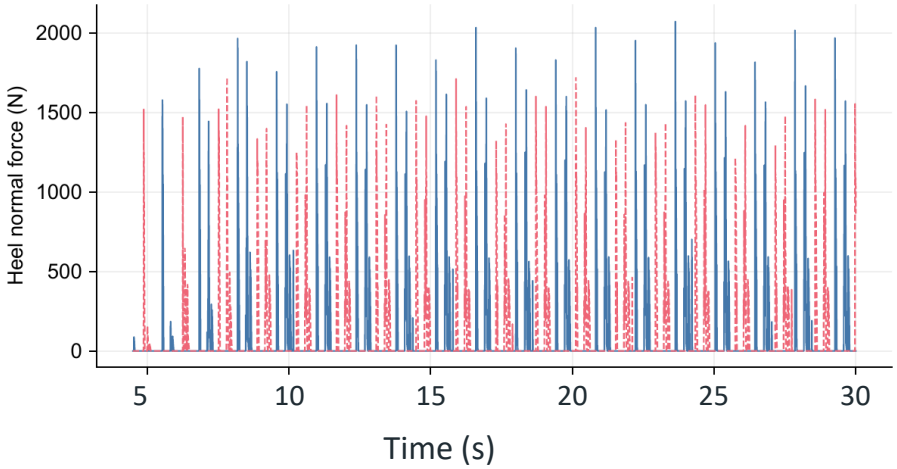


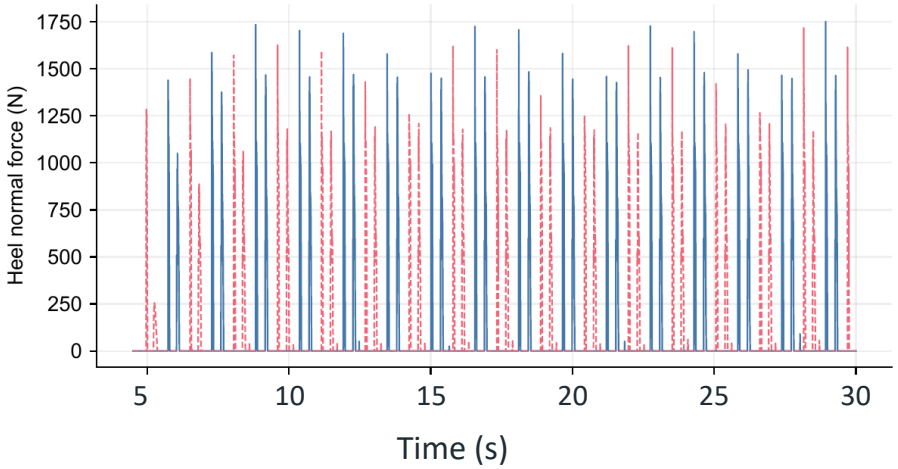
Native recording pending

1 · W2L walker and controller · Dissertation Table 6.1

Sensory feedback changes bilateral foot-loading timing

Left leg: solid · Right leg: dashed



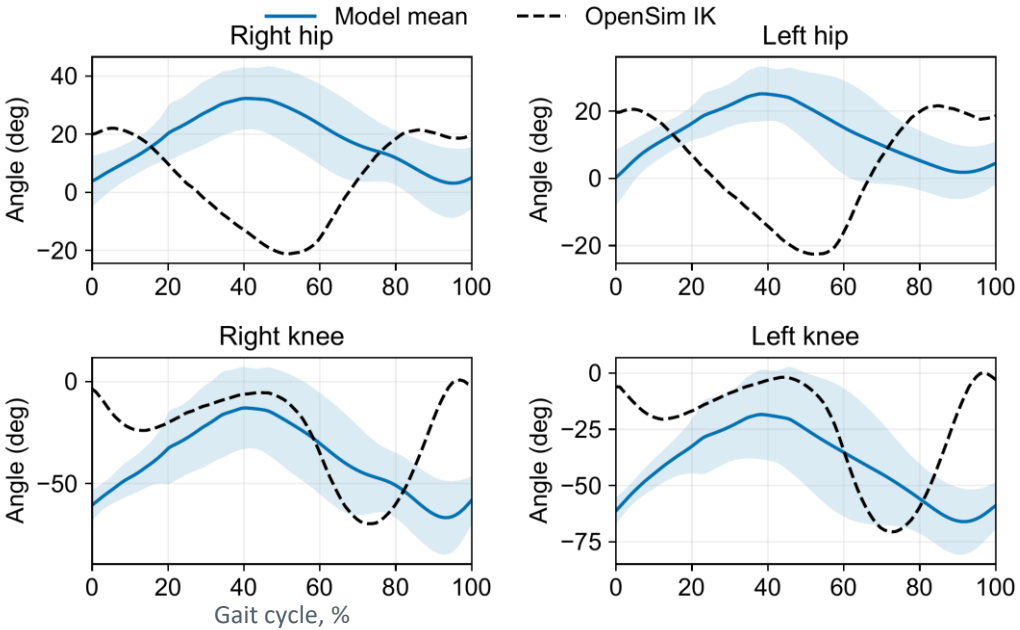


Feedback intact: 0.498 cycle

Feedback disabled: 0.223 cycle

1 · W2L walker and controller · Dissertation Table 6.1

Rhythmic Gait2392 cycles retain hip and knee errors



Hip timing differs

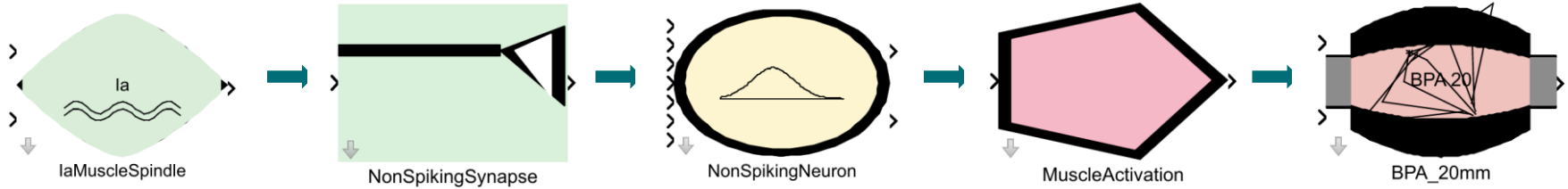
Knee timing differs

No fitted phase shift or amplitude scaling

10 cycles per leg · stance fractions 0.71 / 0.65 versus reference 0.61

2 · Gait2392-based models · Dissertation Table 6.1

Graphical blocks close the loop between muscle mechanics and neurons

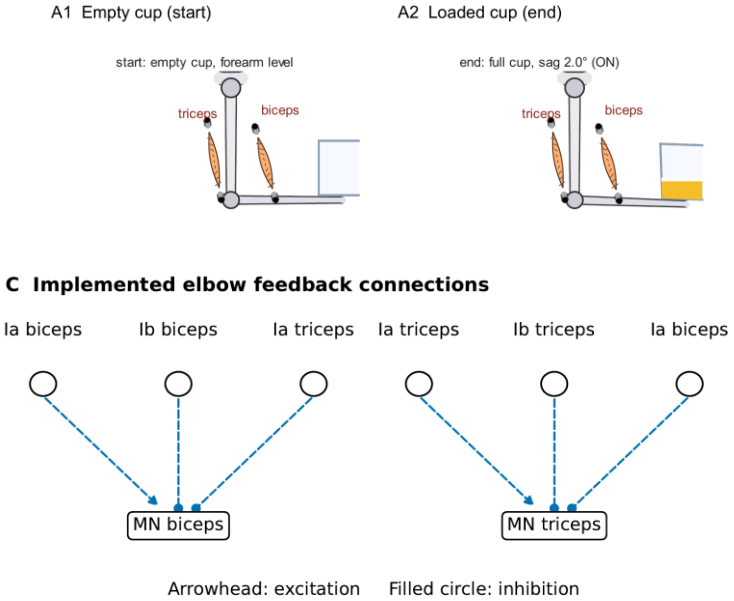


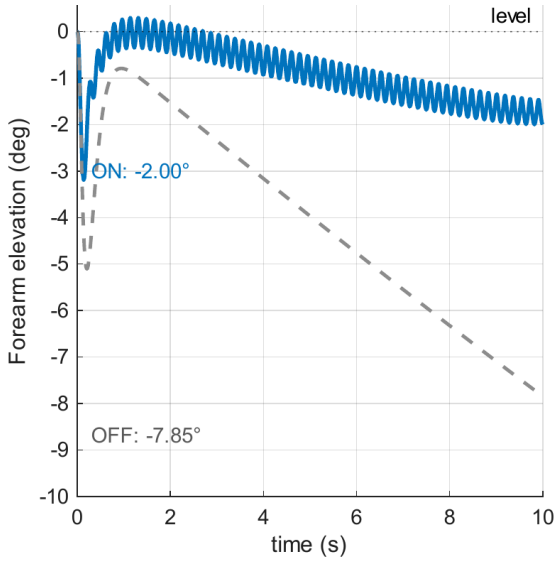
Ia feedback → **Synaptic current** → **Motoneuron** → **Activation** → **BPA force**

Muscle length and velocity return through the coupled model

3 · SNS-Simscape blocks and editor · Dissertation Table 6.1

Sensory feedback reduces sag in the loaded-arm model

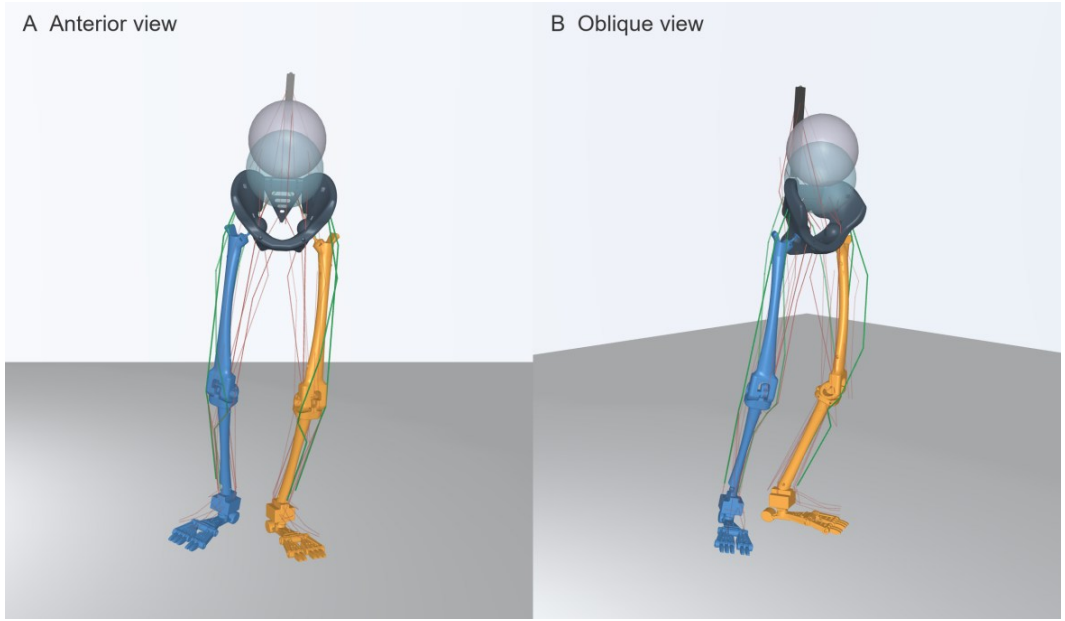




Sag: 2.00° with feedback · 7.85° without feedback

3 · SNS-Simscape blocks and editor · Dissertation Table 6.1

Extra abductor control prevents crossing in the AH001 test



15–40 s	Off	On
Crossed states	67%	0%
Min. spacing	−38.3 mm	142.4 mm

54 reference-tension actuators; 80% vertical + lateral/angular assistance
MN clamp cancels PF/reflex currents; regular stepping unverified

4 · CAD-derived robot model · Dissertation Table 6.1

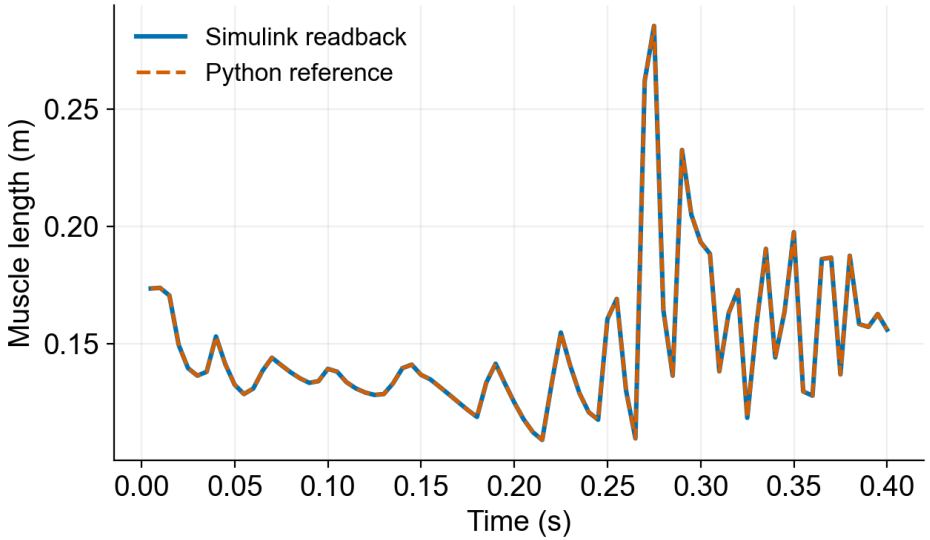
The MuJoCo–Simulink bridge preserves measured signals

SNS-Simscape controller

Activation commands

Sensory measurements

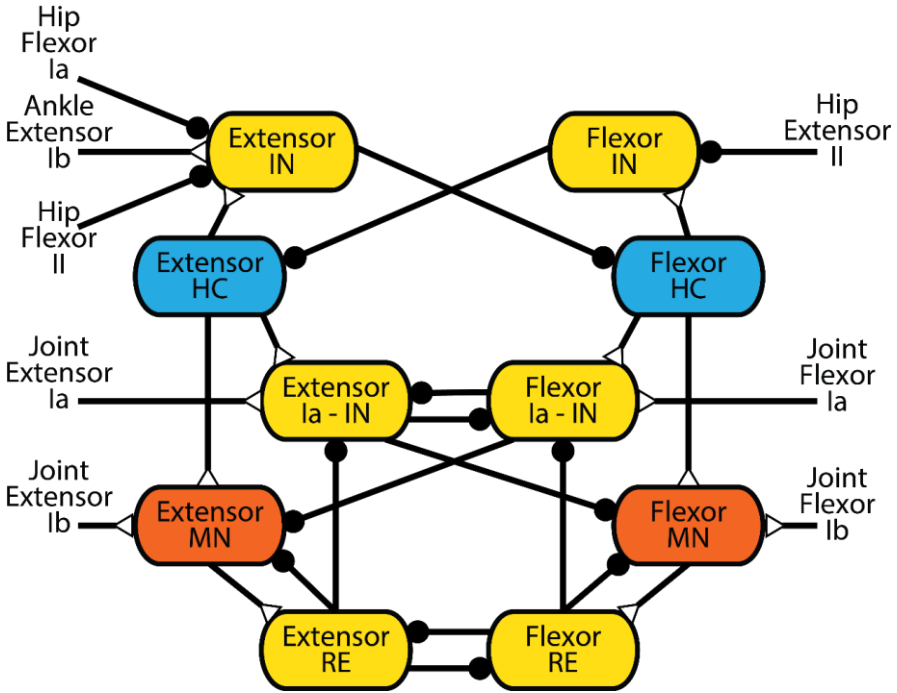
MuJoCo mechanical model



5 sensor channels · 80 scored steps · maximum deviation 0

5 · MuJoCo–Simulink integration · Dissertation Table 6.1

The tools make each of Hunt's four training steps observable



1. **Set intrinsic CPG rhythm**

   SNS blocks and open-loop traces

2. **Train feedback → CPG**

   Afferent playback and timing

3. **Train CPG/PF → MN**

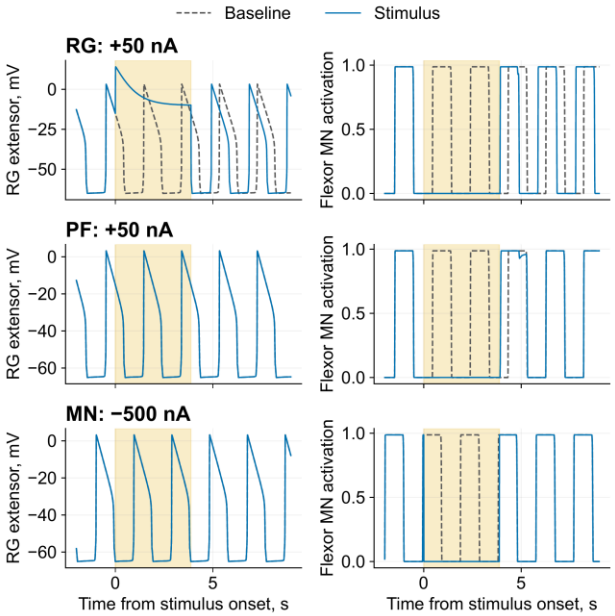
   Motor-neuron peak comparisons

4. **Refine Ia/Ib → MN**

   Closed-loop pathway tests

6 · Curriculum and experiment drivers · Dissertation Table 6.1

Two deleted bursts can preserve or reset the rhythm



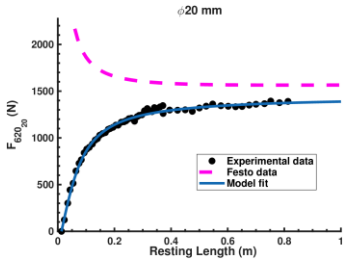
Stimulated layer	RG shift, cycles
RG	0.211
PF	0.000
MN	0.000

RG: rhythm generator
PF: pattern formation
MN: motor neuron

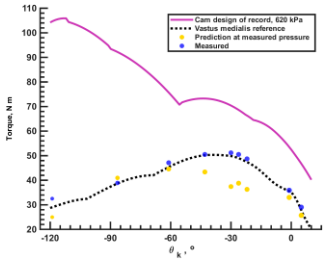
Reduced Deng-derived circuit · two baseline periods · suspended model

7 · Literature circuits and variants · Dissertation Table 6.1

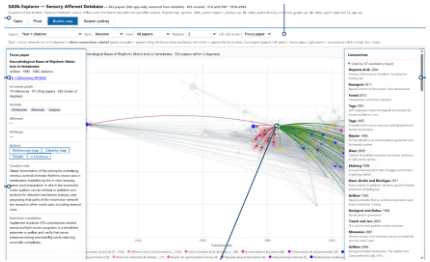
Researchers can connect source evidence to a tested actuator or pathway



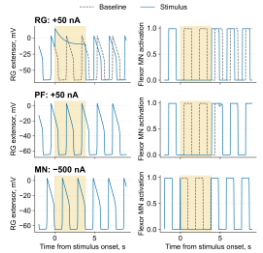
Select an actuator



Test a route



Trace a sensory hypothesis



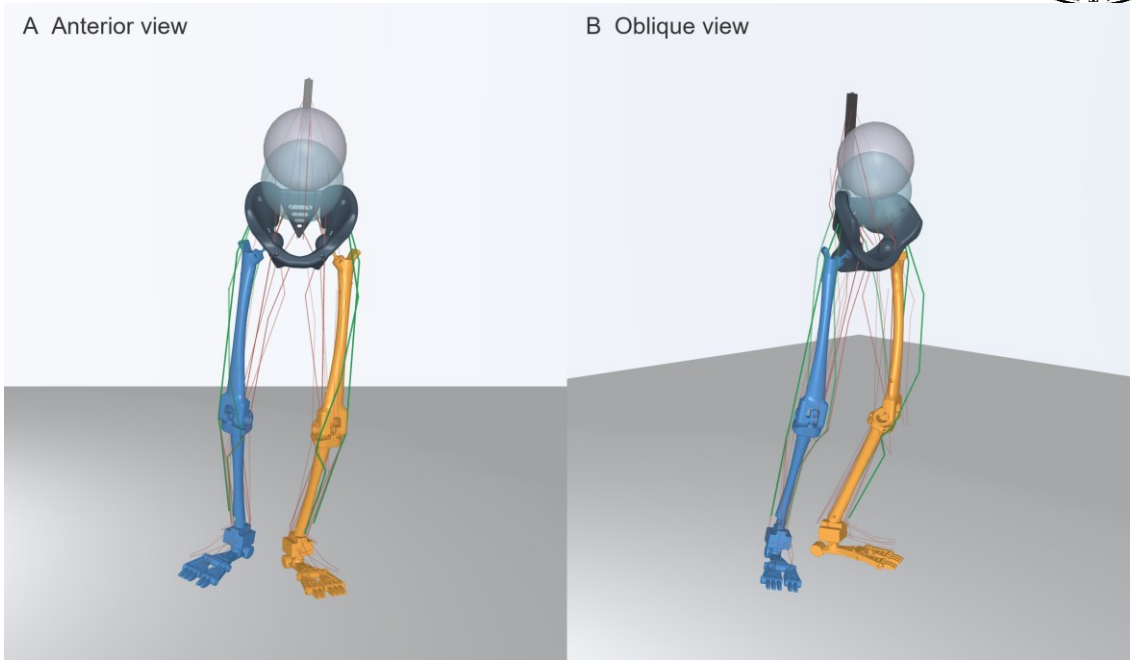
Test a pathway

Dissertation Chs. 3–7

The next experiments must transfer the models to loaded robot movement

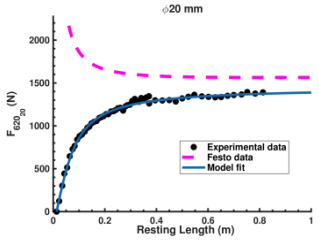


Dynamic loading and physical transfer

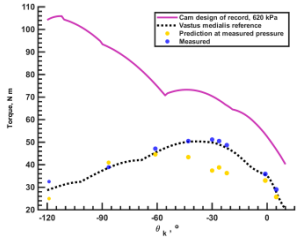


Graded support reduction

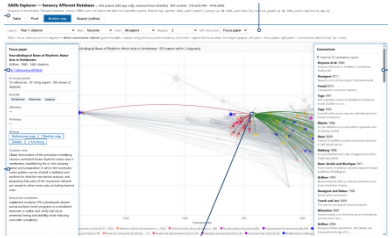
The dissertation makes actuator, route, and feedback choices testable



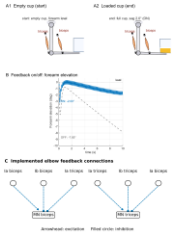
Predict force



Meet selected torque targets



Ground sensory hypotheses

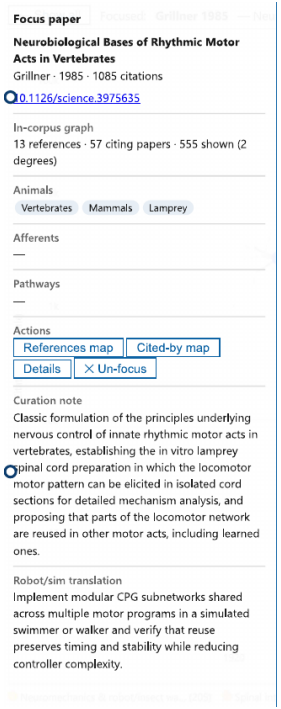


Test neural mechanisms

Thank you · Questions

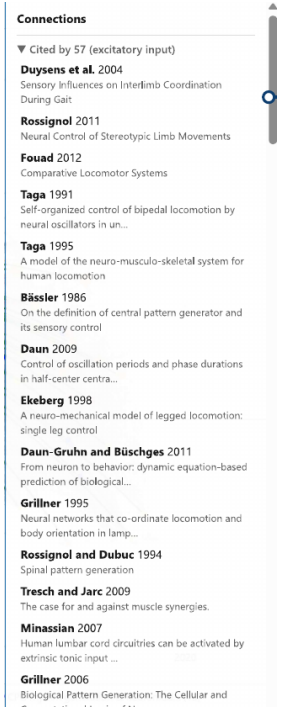
Dissertation Conclusion

SADb keeps experimental context beside the proposed robot interpretation



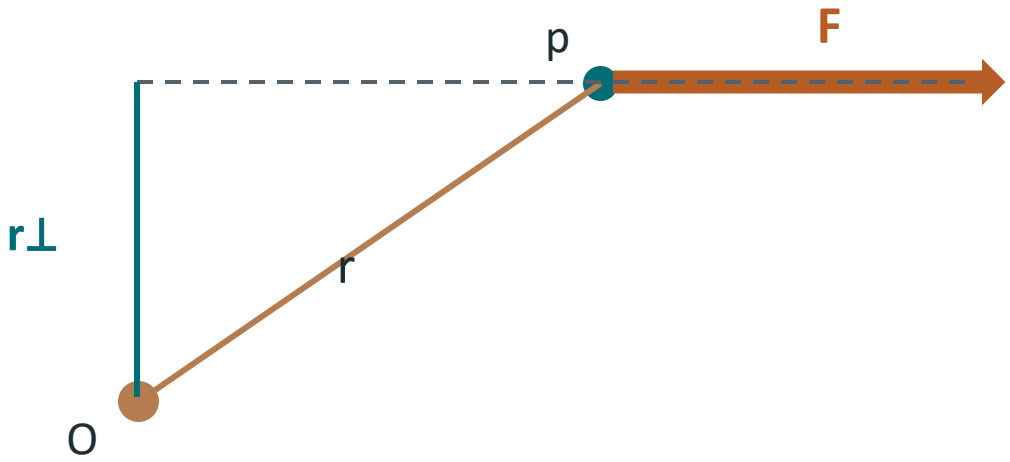
Source and experimental context

Robot interpretation to test



Dissertation Ch. 5 · Focus record and connected sources

Joint torque depends on force direction and moment arm



$$\tau_z = (r \times F) \cdot \hat{z}$$

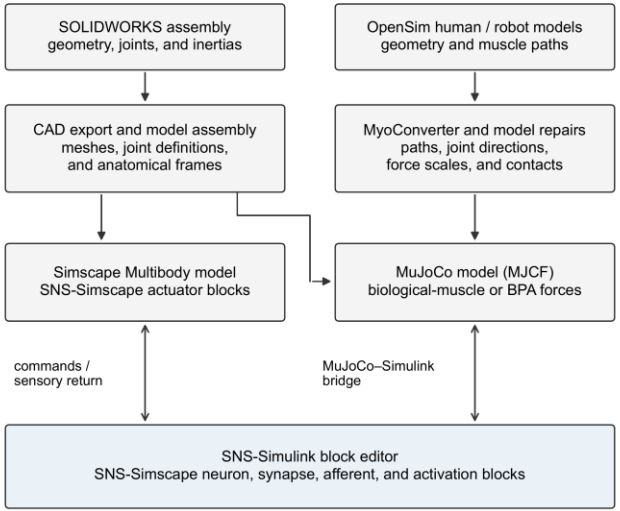
$$|\tau_z| = F\, r_\perp$$

Routing also changes actuator length

A force model needs a transmission model to predict delivered torque

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

The model workflow supports distinct mechanical implementations



Command exchange, circuit agreement, and motion each require their own check

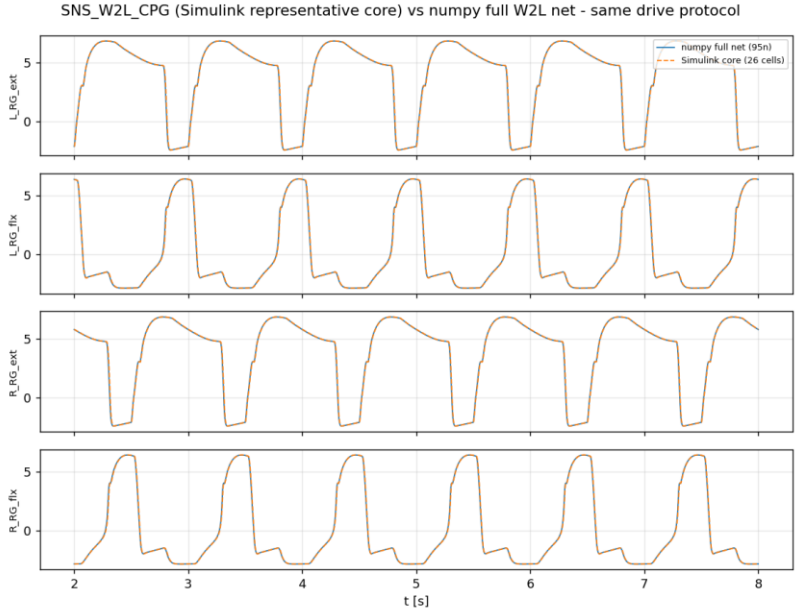
Dissertation Ch. 6, Model construction and neural-controller interfaces

Staged tests separate rhythm, feedback, and mechanical support

Condition introduced	Question tested
Suspended, without sensory feedback	Can the circuit generate activity?
Sensory feedback	How does feedback reshape the output?
Standing and perturbations	Can the model maintain posture?
Supported ground contact	Does physical coordination meet the criterion?
Reference and intervention comparisons	Which mechanism or mismatch explains the result?

Dissertation Ch. 6, curriculum Table 6.2

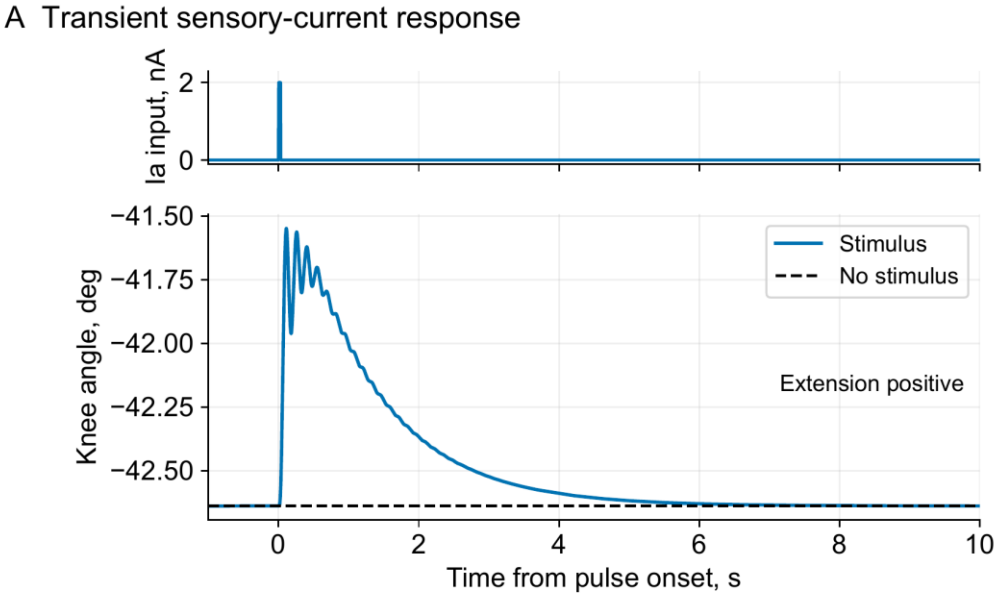
Matched rhythm cores reproduce the same driven bursts

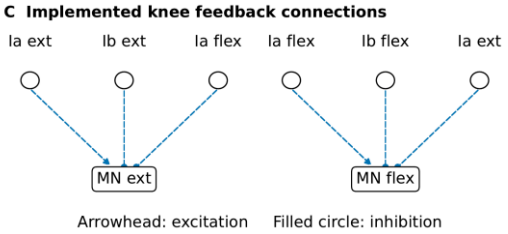


1.000 s period · 10 expected bursts · correlation ≥ 0.99996

Dissertation Ch. 6 and Appendix D · Current saved verification records

The knee model produces a transient sensory response





2 nA for 30 ms

1.09° at 111 ms

Recovery at 6.355 s

Six neural saturation widths changed in this configuration

Dissertation Ch. 6 and Appendix D · Current saved verification records

The evidence establishes specific capabilities within clear limits

Evidence level	What the result establishes	Remaining boundary
Measured hardware	Isometric torque capacity	Dynamic response and transfer
Neural demonstrations	Specified response and intervention effect	Circuit- and protocol-specific
Supported W2L / human models	Coupling and coordination tests	External support remains
AH001 integration	Hybrid allocation; PF/reflex currents cancelled	80% vertical + lateral/angular support

AH001: reference tensions, MN clamp, provisional feet, and ideal brackets

No harness-free walking has been demonstrated

Current manuscript Ch. 6 / Appendix D · Session 1 AH001 records

The adopted identification reserves tests in each configuration

Configuration	Training	Held out	Not used
Pinned flexor	2	2	1
Pinned extensor	3	3	3

Shared effective stiffness pair: $\chi_1 = 39.98$, $\chi_2 = 14.73$ kN/m

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

Physical dimensions differ from prediction settings

Actuator	Resting length, mm	Tendon length, mm
Flexor 1	561	28
Flexor 2	560	30
Extensor 1	828	92
Extensor 2	826	85

Prediction inputs: flexor 561.49 / 5.00 mm; extensor 827.74 / ≈92.49 mm

Torque manuscript, Physical Tests · Dissertation Appendix C dimensions

Stimulation duration changes the deletion interpretation

Protocol	RG shift, cycles	Motor shift, cycles
RG, two periods	0.211	0.211
PF, two periods	0.000	0.000
MN, two periods	0.000	0.000
PF, one second	0.000	0.208

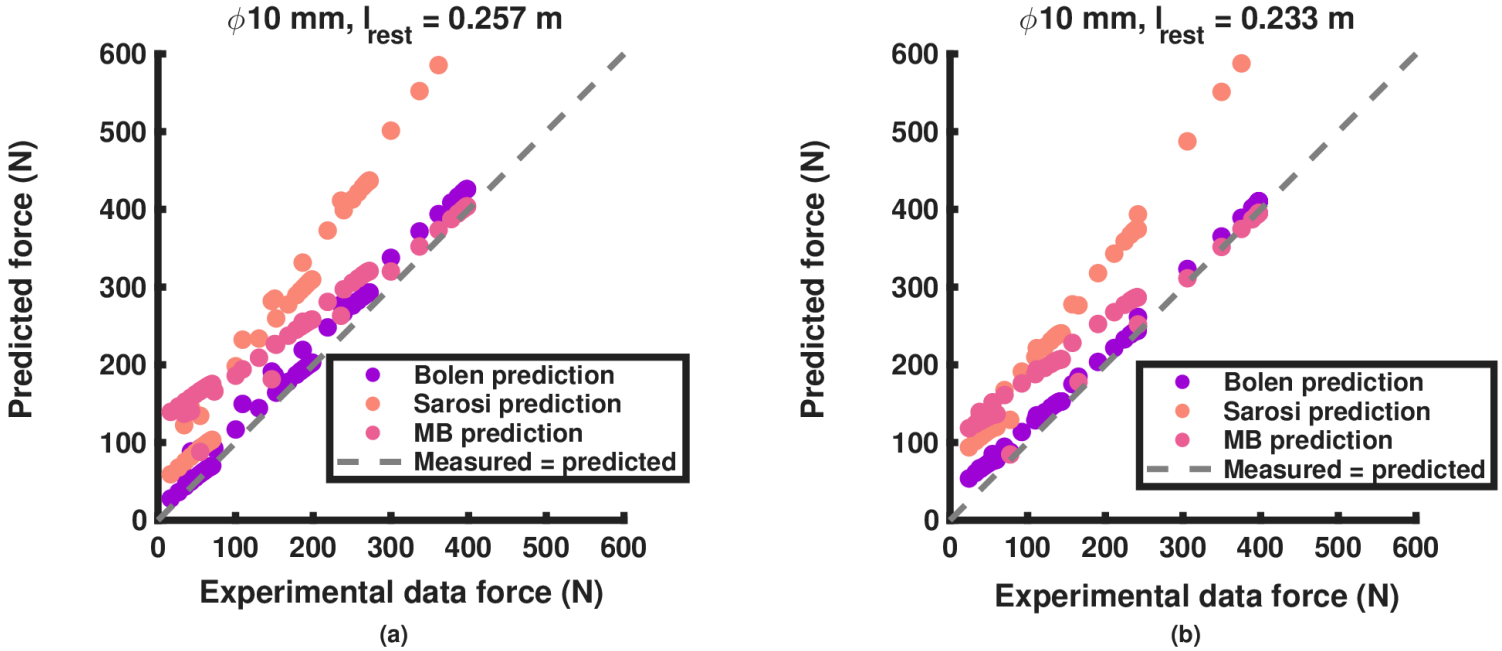
stimulus_results_all.csv · Current Appendix D layer-stimulus protocols

AH001 separation control changes a matched diagnostic

15–40 s measure	Control off	Control on
Reversed ankle ordering	67%	0%
Minimum separation	−38.3 mm	142.4 mm
Right TFL mean activation	22.3%	81.8%
Vertical assistance	80% BW	80% BW
Sampled opposite-leg penetration	2.700 mm	0 mm

AH001_20261028/PLACEMENT_RECOMMENDATION.md · asset_provenance.json

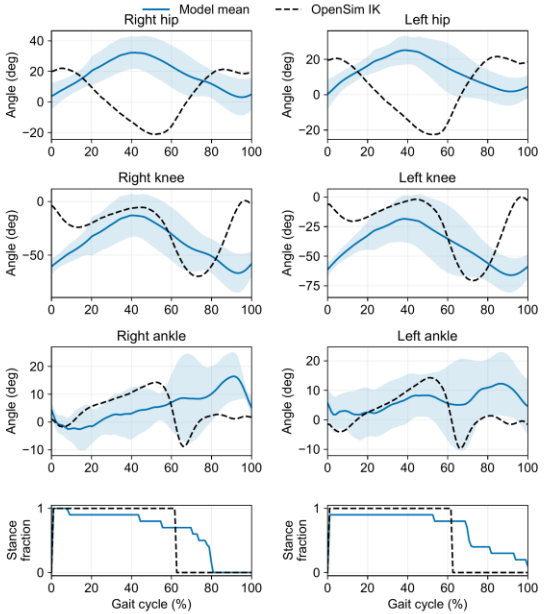
Reported force comparisons favor the model in the shown cases



Measured-versus-predicted force at the two displayed actuator lengths

Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

The complete IK overlay also exposes contact differences

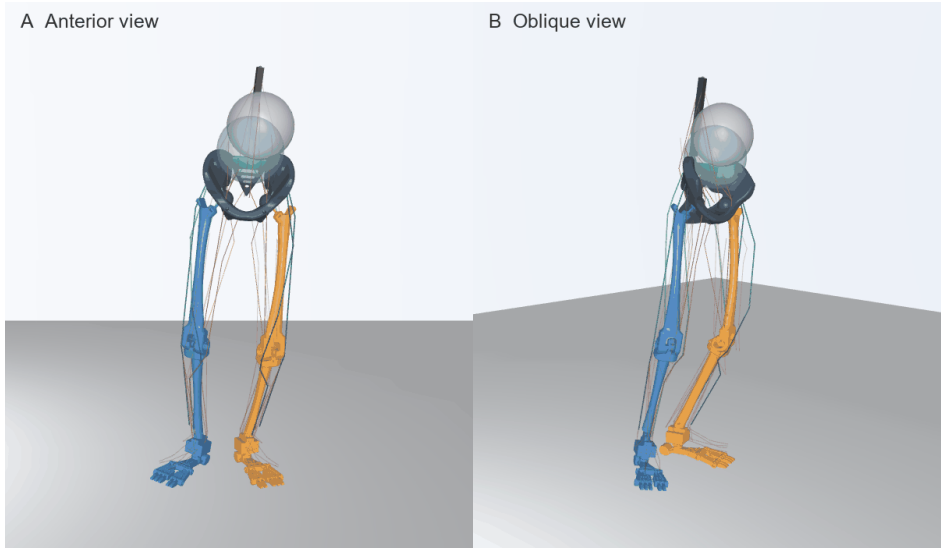


Model mean and standard deviation · OpenSim IK reference

Dissertation Ch. 6 and Appendix D · Current saved verification records

The saved AH001 replay shows supported hybrid motion

Supported CAD robot with increased abductor control



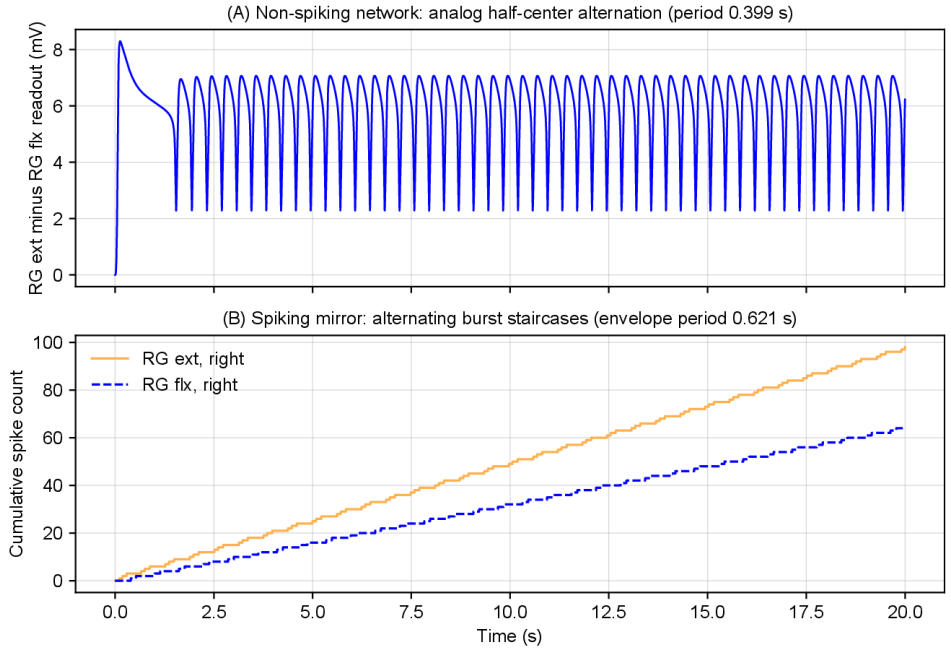
t = 20.00 s | vertical support 80% BW | minimum activation 5%

Hybrid control; reference tensions; lateral/angular assistance and knee-stop surrogates

Optional 12-second replay · same supported-control diagnostic

AH001_20261028/asset_provenance.json · saved 20–32 s interval

Changing neural encoding changes rhythm timing



0.399 s non-spiking → 0.621 s spiking · 55.6% longer

7 · Literature circuits and variants · Dissertation Table 6.1